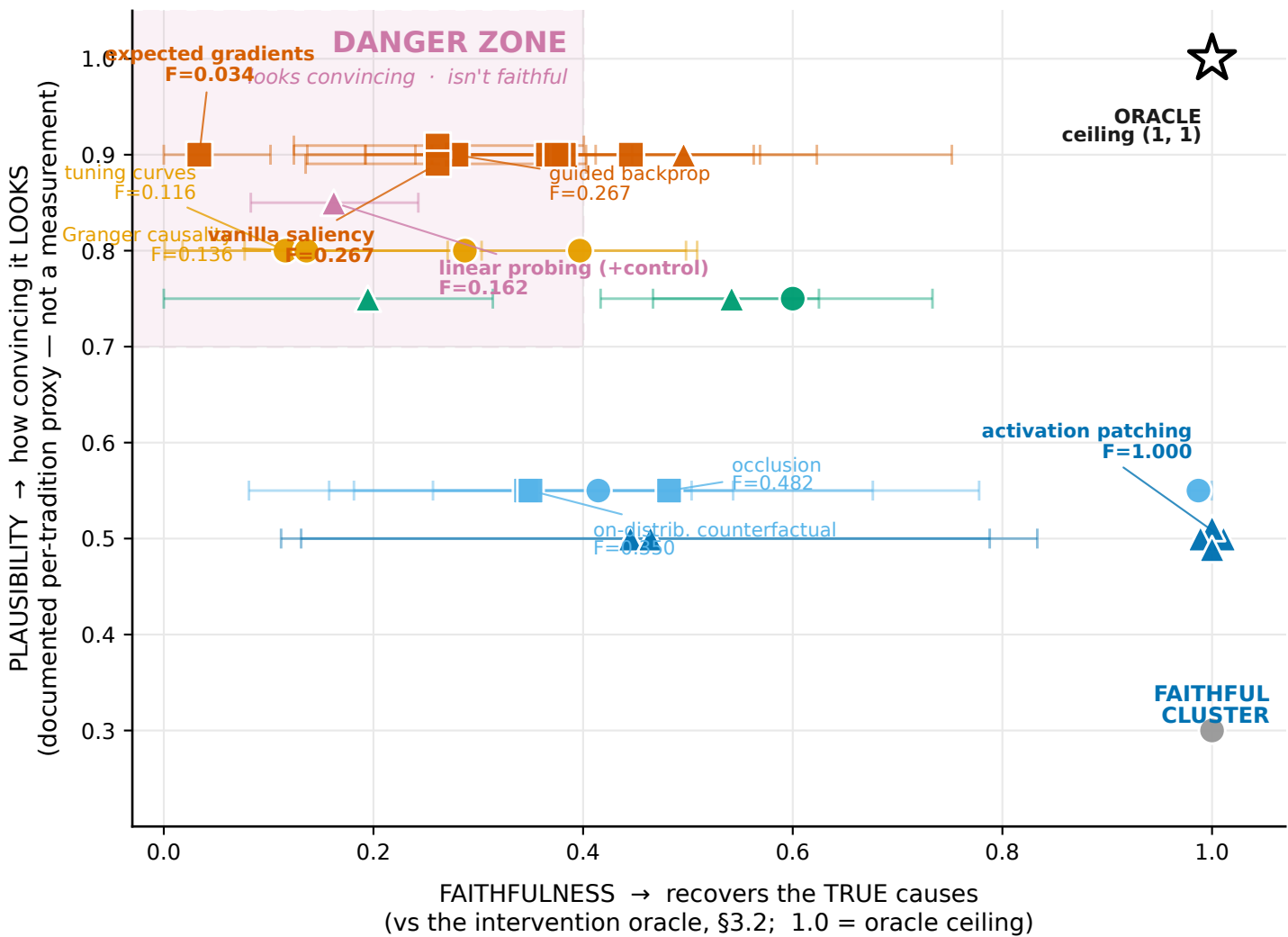


# Plausible ≠ faithful: every interpretability method scored where the truth is known exactly

All 31 rows (30 methods + the oracle positive control) across the three traditions, vs the intervention oracle — the most plausible-looking methods are the least faithful.



experimental battery (marker) + uncertainty

● Phase A — neuroscience battery (Kording)

■ Phase B — attribution / XAI

▲ Phase C — mechanistic interpretability

★ oracle ceiling

— 95% CI (bootstrap over games)

tradition (colour)

● causal mediation / patching

● intervention / occlusion

● gradient / saliency

● correlational (tuning, Granger, ...)

● linear probing

● dim-reduction (NMF/PCA/SAE)

● descriptive baseline

## KEY RESULTS

every number reads from a committed record

| finding          | number            | interpretation                                               |
|------------------|-------------------|--------------------------------------------------------------|
| Faithful cluster | $\bar{F} = 0.681$ | causal / intervention methods recover the true causes        |
| Danger zone      | $\bar{F} = 0.294$ | gradient / correlational: familiar yet least faithful        |
| Big contrast     | 0.27 → 1.00       | vanilla saliency vs activation patching (default → faithful) |
| Worst offender   | $F = 0.034$       | expected gradients: high plausibility, near-zero F           |
| Oracle control   | $F = 1.000$       | intervention oracle anchors the ceiling at (1, 1)            |

Means: causal/intervention vs gradient/correlational families; contrast & worst-offender numbers cross-checked against the Phase-A/B/C records (see Supplement number-provenance table).